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Financial Literacy and Health Insurance Uptake: Evidence From the Households in Tanzania

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Abstract: This study examines the relationship between household financial literacy and health insurance consumption in Tanzania using nationally representative data from the FinScope Tanzania 2023 Survey. Using a probit model and an instrumental variable approach to address endogeneity, we find that financial literacy, as measured by numeracy and borrowing skills, significantly increases the likelihood of health insurance enrollment. The analysis reveals substantial disparities across socio-demographic groups, with stronger effects observed in urban areas, among employed individuals, and younger heads of household (Hhs). Mechanism analyses suggest that financial literacy improves understanding of insurance functionality and reduces cost-related barriers, though it may lower perceived benefits due to more critical evaluation of plans. With only 11.9% of households insured, our findings underscore the urgent need for integrated financial education and insurance literacy programs to enhance coverage, reduce out-of-pocket health expenditures, and advance progress toward universal health coverage in Tanzania.

Keywords: Financial literacy; Health insurance consumption; Financial education; Tanzania

1. Introduction

Health insurance is a key component of global healthcare systems, providing broad financial protection and access to healthcare services. In developing countries, however, health insurance coverage remains low, and many individuals and households face significant barriers to enrolment. For instance, in many African countries, including Tanzania, the health insurance enrolment rate is reported to be below 10 (Chemouni, 2018). Many countries have begun introducing health insurance for people experiencing poverty because it allows households to prepay for healthcare, reducing the share of catastrophic health expenditures they must pay out of pocket (Geng et al., 2018). Low health insurance penetration in these regions often stems from limited awareness, affordability issues, and cultural attitudes toward insurance (Bayked et al., 2023). Low awareness and limited understanding of health insurance among individuals threaten uptake in developing countries (Sekhri & Savedoff, 2005).

The Tanzanian health system has to confront significant challenges in achieving universal health coverage. Many households there rely on out-of-pocket payments, leading to financial hardship and inequity in accessing healthcare services (Binyaruka et al., 2021; Financial Sector Deepening Tanzania, 2023b). The Community Health Fund (CHF) and the National Health Insurance Fund (NHIF) together play a pivotal role in providing affordable health insurance. However, their coverage is limited, particularly among the informal sector and rural populations (Durizzo et al., 2022). Financial literacy is essential for shaping household decisions about health insurance by improving understanding of insurance products, strengthening risk management, and fostering all-inclusive financial planning. For instance, individuals with higher financial literacy are more likely to comprehend the benefits and terms of insurance policies, resulting in increased uptake (Lusardi & Mitchell, 2014).

From a governance and risk-management perspective, health insurance is a formal household risk-pooling mechanism that converts uncertain and potentially catastrophic medical expenditure into a more predictable premium. This function is especially significant wherever out-of-pocket payment could compel households to

borrow, sell assets, reduce essential consumption, or forgo care. Evidence from low- and middle-income countries nevertheless showed that insurance does not provide uniform financial protection; concomitant outcomes depend on scheme design, benefit coverage, provider's access, administrative capacity, and accountability (Eze et al., 2023; Rahman et al., 2022). Accordingly, financial literacy matters not only for enrolment but also for household risk governance because it may help households assess coverage, costs, exclusions, and the reliability of available schemes.

Several studies have examined the relationship between financial literacy and health insurance uptake, primarily in developed countries. For example, Clark et al. (2014) found that higher financial literacy correlated with greater health insurance coverage among U.S. adults. A cross-country study by Zheng et al. (2025) recommended exploring underrepresented regions and emerging trends, with implications for expanding health insurance coverage and promoting healthcare equity. The role of health insurance in improving household healthcare utilization is widely debated in many developing countries because poverty, high user fees, and out-of-pocket payments strain household incomes (Kitole et al., 2023).

Household financial literacy is critical for users to make informed financial decisions, including health insurance consumption. In Tanzania, where health insurance coverage is relatively low, knowing the role of financial literacy could inform efforts to improve health insurance uptake and strengthen households' financial risk protection (Financial Sector Deepening Tanzania, 2023b; Lotto, 2020a). Despite evidence from developed countries, causal understanding of how financial literacy influences health insurance uptake in Sub-Saharan Africa remains limited, particularly in Tanzania. This paper examined how two dimensions of financial literacy, i.e., numeracy and borrowing skills, were related to health insurance enrolment in Tanzania, encompassing heterogeneity by rural–urban residence, employment status, and age.

The structure of this paper is organized as follows: Section 2 reviews the literature. Section 3 presents the methodology, including data sources, key variables, and models. Section 4 discusses the empirical results and further analyses. Finally, we conclude the paper with insights into future research directions.

2. Literature Review

2.1 Health Insurance in Tanzania: Coverage and Challenges

The CHF was introduced to provide health insurance to rural populations and workers in the informal sector, while the NHIF targets formal-sector employees. Despite these initiatives, enrolment rates remain low, with significant disparities in coverage between urban and rural areas (Borghi et al., 2013). The government's efforts to expand health insurance coverage, such as the CHF and the NHIF, have had limited success, thus highlighting the need for complementary strategies to enhance uptake (Borghi et al., 2013; Macha et al., 2014).

Tanzania advocated community-based health insurance (CBHI) in the 1990s as part of the broader health sector reforms to improve healthcare financing and increase access to health services. Two main schemes were established: the CHF, which targets rural households and the informal sector, and Tiba Kwa Kadi (TIKA), which focuses on individuals in urban areas. Both schemes receive central government support through matching grants. However, challenges such as low enrolment and weak management propel the government to transfer CHF administration to the NHIF in 2009 in order to enhance operational efficiency and expand coverage. Subsequent to this transition, the NHIF implemented several reforms, including improvements to the benefit package and administrative systems. Nevertheless, low participation in CBHI schemes presents a persistent challenge in Tanzania, as well as in many other low- and middle-income countries (Kigume & Maluka, 2021).

2.2 Financial Literacy: Concept, Measurement, and Global Evidence

The Organisation for Economic Co-operation and Development (OECD) defined financial literacy as knowledge and understanding of financial concepts and risks; it included the skills, motivation, and confidence to apply such knowledge and understanding to make effective decisions across a range of financial contexts, to improve the financial well-being of individuals and society, and to encourage participation in economic life (Lusardi, 2019). Meanwhile, Lusardi & Mitchell (2014) argued that financial literacy predominantly affected individuals' competence to manage finances, save, invest, and borrow effectively.

2.3 Financial Literacy and Health Insurance Uptake

Financial literacy is prevailingly recognized as a key determinant of financial behaviour and risk-management choices (Lusardi & Mitchell, 2014). It could elevate households' ability to compare premiums and benefits, interpret exclusions, understand risk pooling, and assess insurance's value in the long term. Several studies have linked financial literacy to health-related financial behaviours. Evidence from China demonstrated that higher financial literacy was positively associated with both the probability of holding life insurance and the premiums

paid, thus suggesting that financial knowledge could demolish information barriers in making complex insurance decisions (Wang et al., 2021). Likewise, household evidence indicated that financial literacy could reduce poverty vulnerability partly by encouraging participation in commercial insurance and improving the choice of formal financial channels (Wang et al., 2022).

Previous evidence collected from developing and low-income economies, however, is not uniformly positive. In Senegal, Bonan et al. (2017) discovered that a standalone insurance-literacy module did not prominently increase health-microinsurance take-up. In contrast, interventions that cut down on entry costs substantially raised adoption, especially among poorer households. This finding suggested that knowledge might be necessary but insufficient when liquidity constraints, trust, product quality, and administrative barriers remained binding. In urban Ethiopia, education, knowledge of the benefit package, and the ability to mobilize funds were associated with willingness to join CBHI. However, affordability was considered to be a major constraint (Deksisa et al., 2020). Recent Ugandan evidence further illustrated that digital literacy was positively associated with insurance inclusion and operated partly through the adoption of insurance technology, emphasizing the growing importance of digital capability in low-penetration insurance markets (Kiwanuka & Sibindi, 2024).

The broader household-risk literature clarified the reasons for promoting insurance participation. A systematic review and meta-analysis encompassing low- and middle-income countries reported that CBHI generally increased healthcare utilization and reduced out-of-pocket and catastrophic expenditure. Nevertheless, the strength of financial protection varied by context and scheme design (Eze et al., 2023). A related scoping review concluded that findings on financial risk protection remained geographically concentrated and often measured only catastrophic expenditure or impoverishment, leaving gaps on coping strategies, forgone care, and low-income-country settings (Rahman et al., 2022). Thus, existing research supports a relationship among literacy, insurance participation, and household resilience, but it also confirms that translating knowledge into enrolment depends on household resources and institutional conditions. Financial literacy positively impacts purchase intentions, and health insurance literacy positively influences intentions. Research on geographical areas remains concentrated in major economies like China, the United States, and India, while smaller or lower-income countries receive insufficient attention (Zheng et al., 2025).

The growing literature satisfactorily addressed three research questions. First, many studies relied on evidence from life insurance, microinsurance, or commercial insurance outside Tanzania. In contrast, comparatively few studies examined household health-insurance enrolment using nationally representative evidence from a low-income East African country. Second, previous research often used a single composite variable to measure financial literacy and may have overlooked that numeracy and borrowing skills influenced insurance enrolment differently. Last, average effects might conceal inequalities in access, since rural location, employment arrangement, and age affected access to providers, income stability, exposure to employment-based insurance schemes, and the ability to use financial education. Tanzania is unique because health insurance coverage remains limited and divided into formal and community-based schemes, where people in rural areas or those informally employed face particular challenges in accessing and affording insurance (Durizzo et al., 2022; Kigume & Maluka, 2021). This research will contribute to the literature by exploring how numeracy and borrowing skills influence insurance enrolment separately.

3. Methodology

3.1 Sources of Data

The study employed a cross-sectional analysis using data from the FinScope Tanzania 2023 Survey (Financial Sector Deepening Tanzania, 2023a). This survey was a nationally representative demand-side survey of adults (16 years or above) living in Tanzania. The survey was conducted through public-private sector collaboration with the Ministry of Finance and Planning, Tanzania and Zanzibar; the Bank of Tanzania; the Financial Sector Deepening Tanzania; the National Bureau of Statistics; and the Office of the Chief Government Statistician, Zanzibar. The survey targeted 10,005 individuals and achieved a 99% response rate, with 9,915 interviews. The survey covered 30 regions in mainland Tanzania and Zanzibar. This study was conducted at the household level; the data were collapsed to 5,298 households, restricting respondents to heads of household (Hhs) with knowledge of health insurance. The data obtained include household demographics, financial behaviour, and health insurance status.

3.2 Description and Measurement of Variables

To measure financial literacy, this paper adopted three standard sets of questions found in numerous surveys in Tanzania and other countries. The questions were based on two concepts: numeracy and borrowing skills, including the capacity to calculate interest rates and understand risk diversification (Lusardi & Mitchell, 2014). Lotto (2020b) also measured financial literacy using interest rates, discounting, and borrowing. This study was based on a two-question concept, in which households were interviewed with questions to test their financial

literacy, including numeracy and borrowing. Table 1 below describes the variables in this study.

Table 1. Description of variables

Variable	Description	Nature
Health insurance	Indicates whether the heads of household (Hhs) is enrolled in health insurance	Dummy variable: It takes the value of 1 if the Hh is enrolled in health insurance, and 0, otherwise.
Financial literacy	Numeracy skills	It is an index obtained through principal component analysis (PCA) of four variables: addition, subtraction, multiplication, and division.
	Borrowing skills	Dummy variable: The variable takes the value 1 if the household shows financial literacy in terms of the capacity to do calculations related to interest rates and an understanding of risk diversification; otherwise, the variable takes the value 0.
Age	Log of the age of the Hh	Continuous variable.
Gender	Sex of the Hh	Dummy variable: 1 if the Hh is male and 0 if female.
Income	Represents annual household income (log)	Continuous variable.
Education level	Represents the highest level of education attained by the Hh, at least secondary school level	Dummy variable: Take the value 1 if the Hh has at least a secondary education, and 0, if otherwise.
Employment status	Represents the employment status of the Hh, whether employed or not	Dummy variable: Take the value 1 if the Hh is employed and 0, if otherwise.
Marital status	Represents the marital status of the Hh, whether he/she is married or not	Dummy variable: 1 if the Hh is married and 0, if otherwise.
Location	This denotes the Hh's place of residence.	Dummy variable: 1 for urban and 0 for rural.

Source: Financial Sector Deepening Tanzania (2023a).

3.3 Empirical Strategy

This study investigated the relationship between household financial literacy and health insurance consumption in Tanzania using a probit model. The probit model suits binary dependent variables, as it assumes a standard normal cumulative distribution function for the probability that the dependent variable is a binary indicator of health insurance enrolment (1 if the household is enrolled in any health insurance scheme, 0 otherwise).

The specific probit model for this study is written as:

$$pr(Y_{\mathrm{i}} = 1) = \beta_0 + \beta_1 \mathrm{FL}_i + \beta_2 X_i + Region_i + \varepsilon_i \quad (1)$$

where, Y_i represents the health insurance status of Hh i(1 if enrolled in health insurance, 0 if not enrolled on the health insurance). FL_i: Financial literacy score of Hh i, X_i is a set of control variables, including household education level, gender, location, income, employment, and age. ε_i is the error term. The coefficient for financial literacy (FL_i) indicates how changes in financial literacy score affect the probability of being insured. These coefficients represent the effect of the respective explanatory variables (age, gender, education, income, employment status, and urban/rural residence) on the probability of being insured.

Table 2. Summary statistics

Variable	Observations	Mean	Standard Deviation	Min	Max
Health insurance	5,298	0.119	0.324	0	1
Numeracy skills	5,298	0.021	1.518	-4.425	1.208
Borrowing skills	5,298	0.096	0.295	0	1
Age	5,298	3.766	0.356	2.773	4.605
Marital status	5,298	0.592	0.491	0	1
Gender	5,298	0.629	0.483	0	1
Education level	5,298	0.207	0.405	0	1
Employment status	5,298	0.092	0.289	0	1
Income	5,298	13.202	1.321	9.903	15.607
Location	5,298	0.336	0.472	0	1

Source: Author computation from Financial Sector Deepening Tanzania (2023a).

3.4 Summary Statistics of Variables

Table 2 indicates that 5,298 households are Hhs in the sample. This is the total number of observations used in the analysis. The results indicated that only about 11.9% of households in the sample had health insurance,

suggesting low health insurance coverage in Tanzania compared with other countries like Ghana. Adjei-Mantey & Horioka (2023) identified that only about 40% of the population was enrolled in the National Health Insurance Scheme (NHIS). Numeracy skills, based on principal component analysis (PCA) scores, are relative to a sample mean of zero; a sample mean of 0.021 implies that households in the sample have a very slight positive deviation of 0.021 standard deviations from the population benchmark. Whereas, 9.6% had financial literacy based on borrowing skills, especially knowledge of interest rates. Therefore, the sample's average financial literacy level was quite low. As indicated, 59.2% of Hhs in the sample were married, and 62.9% of Hhs in the sample were male. Approximately 20.7% of Hhs had at least a secondary education, and only about 9.2% were currently employed. This result highlighted the need for financial literacy programs in Tanzania.

4. Results and Discussion

4.1 Model Diagnostic Tests

Multicollinearity Test

The study opted for a probit model to analyze the relationship between health insurance consumption and household financial literacy in Tanzania, while controlling for households' demographic and socio-economic factors, including age, gender, education, income, employment status, and urban/rural residence. Before running probit estimations, the researchers conducted diagnostic tests to check for multicollinearity, including the variance inflation factor (VIF) and the pairwise correlation matrix. As shown in Table 3, the VIF indicated the extent of multicollinearity. The VIF test shows that all variables have VIF values below 10, suggesting that multicollinearity is not a major concern in this model. Similarly, the correlation coefficients are found mostly below 0.3 on average, indicating no serious multicollinearity problems when these variables are included in our estimation together.

Table 3. Multicollinearity test

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	VIF	1/VIF
(1) Health insurance	1.000	–	–	–	–	–	–	–	–	–	–	–
(2) Numeracy skills	0.077*	1.000	–	–	–	–	–	–	–	–	1.23	0.813
(3) Borrowing skills	0.040*	0.097*	1.000	–	–	–	–	–	–	–	1.19	0.846
(4) Age	0.130*	-0.290*	-0.026	1.000	–	–	–	–	–	–	1.18	0.813
(5) Marital status	-0.121	0.190*	0.014	-0.104*	1.000	–	–	–	–	–	1.47	0.682
(6) Gender_Male	-0.023	0.226*	-0.010	-0.134*	0.543*	1.000	–	–	–	–	1.48	0.676
(7) Education level	0.237*	0.288*	0.039*	-0.282*	0.057*	0.103*	1.000	–	–	–	1.34	0.747
(8) Employment status	0.260*	0.153*	0.048*	-0.140*	0.028*	0.074*	0.311*	1.000	–	–	1.15	0.869
(9) Income	0.015	0.019	0.001	0.011	-0.019	0.011	0.004	0.002	1.000		1.01	0.992
(10) Location	-0.064*	-0.051*	0.006	-0.116*	0.049*	0.018	-0.035*	-0.126*	-0.093*	1.000	1.03	0.995

Note: *$p < 0.1$; VIF = variance inflation factor; the en dash (–) indicates not applicable. Source: Author computation from Financial Sector Deepening Tanzania (2023a).

Model Specification Test

We analsyzed whether the model includes all necessary variables and none that are unnecessary. Auxiliary variables were defined as the predicted value (_hat) and squared predicted value (_hatsq). For a model to be specified appropriately, _hat must be statistically significant. Table 4 shows that _hat is statistically significant at the 1% level ($p = 0.000$), confirming that the model is well specified.

Table 4. Model specification test

Health Insurance	Coefficient	Standard Error	z	$P > \|z\|$	95% Confidence Interval	Interval
_hat	1.078	0.088	12.23	0.000	0.905	1.251
_hatsq	0.042	0.042	1.00	0.316	-0.040	0.125
_cons	0.018	0.051	0.35	0.729	-0.084	0.119

Note: Observations = 5,230; Pseudo $R^2 = 0.21$. Source: Author computation from Financial Sector Deepening Tanzania (2023a).

4.2 Results of the Relationship Between Financial Literacy and Household Health Consumption

Table 5 presents results from a probit model examining the relationship between health insurance coverage and financial literacy, alongside other control variables. The analysis revealed a statistically significant positive relationship between financial literacy and health insurance coverage across various model specifications. In column 1, with no basic controls, the results were positive at the 1% significance level, except in column 4, where

borrowing skills is significant at 10%. Specifically, the coefficient on numeracy skills was 0.09, indicating that Hhs with numeracy skills were more likely to enrol in health insurance coverage. As shown in column 2, the coefficient on borrowing skills was 0.221, indicating that Hhs with borrowing skills were more likely to enrol in health insurance coverage.

The analysis included all other control variables; the results were positive at the 1% significance level. Column 3 shows that the coefficient on numeracy skills was 0.09, indicating that Hhs with numeracy skills were more likely to enrol in health insurance coverage, consistent with the result in column 1. In column 4, the coefficient on borrowing skills was 0.158, indicating that Hhs with borrowing skills were more likely to enrol in health insurance coverage. The positive association between health insurance and financial literacy was consistent across all model specifications, even after controlling for other factors. This finding is consistent with Kopplin (2024), who reported that financial literacy significantly increased the probability of holding health insurance and reduced adverse health-related financial behaviours. Similarly, Hubbard (2024) concluded that financial knowledge was associated with a greater likelihood of obtaining health-insurance coverage and a lower likelihood of accumulating medical debt. Evidence from India also concurred that financial literacy was one mechanism through which digital financial participation promoted voluntary health-insurance enrolment (Sengupta & Rooj, 2025). Therefore, the stability of the coefficients across the model specifications suggested that financial literacy was an important correlate of health-insurance enrolment in Tanzania.

4.3 Results of Heterogeneous Analysis

To ensure the robustness of our findings, several heterogeneous analyses were conducted to test the robustness of the findings. The analysis included heterogeneous effects by urban and rural locations, employment status, and age categories.

Table 5. Results from Probit Regression analyzing the relationship between financial literacy and household health consumption

Probit Model	**(1)**	**(2)**	**(3)**	**(4)**
Dependent Variable: Health Insurance				
	0.09***	–	0.09***	–
Numeracy skills	(0.018)	–	(0.020)	–
	–	–	[0.014]	–
	–	0.221***	–	0.158*
Borrowing skills	–	(0.073)	–	(0.081)
	–	–	–	[0.0127]
	–	–	1.481***	1.387***
Age	–	–	(0.093)	(0.089)
	–	–	[0.233]	[0.219]
	–	–	-0.012	0.019
Marital status	–	–	(0.063)	(0.063)
	–	–	[0.0140]	[0.0099]
	–	–	-0.169***	-0.134**
Gender	–	–	(0.064)	(0.063)
	–	–	[-0.027]	[-0.021]
	–	–	0.997***	1.053***
Education level	–	–	(0.066)	(0.065)
	–	–	[0.157]	[0.117]
	–	–	0.942***	0.945***
Employment status	–	–	(0.074)	(0.074)
	–	–	[0.148]	[0.149]
	–	–	0.023	0.023
Income	–	–	(0.02)	(0.019)
	–	–	[0.004]	[0.004]
_cons	-1.107***	-1.121***	-7.414***	-7.116***
	(0.122)	(0.123)	(0.463)	(0.453)
Region dummies	YES	YES	YES	YES
Observations	5,298	5,298	5,230	5,230
Pseudo R^2	0.042	0.036	0.213	0.208

Note: Robust standard errors in parentheses; marginal effects in square brackets. ***, **, * denote significance at 1%, 5%, and 10%, respectively. The en dash (–) indicates not applicable.

4.3.1 Heterogeneous effects by rural and urban location

We examined the robustness of the findings by understanding the relationship between health insurance enrolment and financial literacy separately for urban and rural households, along with other control variables.

Table 6 indicates that numeracy skills are more strongly associated with health insurance enrolment in urban than in rural areas, while borrowing skills are significant only for urban households. This pattern may reflect closer proximity to insurers and health facilities, greater exposure to formal financial products, more reliable information, and more frequent monetary transactions in urban settings. Rural households may understand basic calculations, yet remain unable to translate that knowledge into enrolment because of irregular incomes, distance, limited provider networks, and fewer suitable products. The weaker and less stable association for borrowing skills is also plausible because knowledge of interest and credit does not directly imply an ability to evaluate insurance benefits, exclusions, or provider quality; it may matter only where credit markets and insurance payment options are accessible. These results demonstrated that financial literacy, measured through numeracy skills and borrowing skills, positively correlated with health insurance enrolment. The analysis revealed a distinct disparity between urban and rural areas. Urban settings showed a stronger influence of financial literacy types on health insurance enrolment, particularly numeracy skills, while borrowing knowledge had a smaller and less consistent influence. In contrast, these variables had less impact on rural areas. Socio-economic factors such as age, education level, and employment status are significantly positive predictors of health insurance coverage. Gender differences exist, particularly in rural areas where male Hhs are less likely to enrol in insurance coverage.

Table 6. Heterogeneous effects by rural and urban location

Probit Model	**(1)**	**(2)**	**(3)**	**(4)**
Dependent Variable: Health Insurance				
Financial Literacy Measure	**Numeracy skills**		**Borrowing skills**	
	Urban	**Rural**	**Urban**	**Rural**
Numeracy skills	0.123***	0.047*	–	–
	(0.036)	(0.025)	–	–
	[0.0081]	[0.0030]	–	–
Borrowing skills	–	–	0.23*	0.107
	–	–	(0.118)	(0.114)
	–	–	[0.0271]	[0.0136]
Age	1.638***	1.413***	1.551***	1.357***
	(0.138)	(0.133)	(0.136)	(0.125)
	[0.0280]	[0.0160]	[0.0279]	[0.0151]
Marital status	-0.047	0.031	-0.02	0.048
	(0.091)	(0.09)	(0.091)	(0.09)
	[0.0207]	[0.0107]	[0.0208]	[0.0107]
Gender	-0.10	-0.152*	-0.061	-0.128
	(0.093)	(0.09)	(0.092)	(0.089)
	[0.0211]	[0.0107]	[0.0210]	[0.0106]
Education level	0.959***	0.886***	1.036***	0.91***
	(0.093)	(0.101)	(0.091)	(0.1)
	[0.012]	[0.0120]	[0.0187]	[0.0119]
Employment status	0.686***	1.284***	0.678***	1.294***
	(0.094)	(0.122)	(0.093)	(0.122)
	[0.0204]	[0.0144]	[0.0203]	[0.0144]
Income	0.01	0.031	0.006	0.031
	(0.03)	(0.027)	(0.03)	(0.027)
	[0.0067]	[0.0032]	[0.0068]	[0.0032]
_cons	-7.793***	-7.32***	-7.462***	-7.159***
	(0.71)	(0.643)	(0.708)	(0.623)
Region dummies	YES	YES	YES	YES
Observations	1690	3476	1690	3476
Pseudo R^2	0.201	0.196	0.195	0.195

Note: Probit coefficients with robust standard errors in parentheses; marginal effects in square brackets. Urban–rural differences (Wald tests) are statistically significant. ***, **, * denote significance at 1%, 5%, and 10%, respectively. The en dash (–) indicates not applicable.

4.3.2 Heterogeneous effects by employment status

Table 7 tests robustness by examining the relationship between health insurance enrolment and financial literacy, based on how numeracy and borrowing skills interact with health insurance across Hhs' employment status, along with other control variables. Numeracy skills are shown to be positively related to enrolment across all employment status groups, while borrowing skills do so only for full-time and privately employed Hhs. Numeracy skills refer to the ability to make decisions; they enable premium budgeting, cost/benefit comparison, and understanding contribution schedules regardless of employment status. Borrowing skills, in turn, become relevant under formal borrowing facilities, constant money flows, linkages to payrolls and transactions with banks. Such factors are characteristic of full-time and private rather than part-time and unemployed households. Instability in borrowing skill estimates could also reflect conceptual differences between interest-calculation knowledge and knowledge

about insurance, which involves probability concepts and claim uncertainty. Consequently, programmes should include numeracy and insurance education along with suitable products and contributions schemes for workers with non-standard income. These results showed a stronger role for numeracy skills across all Hhs' employment statuses. This aligns with other studies such as Zheng et al. (2025), which has a more consistent and stronger impact on health insurance enrolment than on borrowing skills. Fully employed households likely benefit most from financial literacy because of structured work benefits. Part-time workers and unemployed individuals benefit less because they have limited access to employer-provided options. Education has consistently boosted health insurance enrolment across groups, thus underscoring its critical role in financial decision-making. Older individuals are more likely to enrol in health insurance, but the effect of age also varies by employment status.

Table 7. Heterogeneous analysis on employment status

Probit Model	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
	Dependent Variable: Health Insurance							
Financial Literacy Measure	**Numeracy Skills**				**Borrowing Skills**			
	Full-time Employees	**Part-time Employees**	**Unemployed**	**Privately Employed**	**Full-time Employees**	**Part-time Employees**	**Unemployed**	**Privately Employed**
Numeracy skills	0.414***	0.086**	0.1**	0.105***	–	–	–	–
	(0.108)	(0.04)	(0.039)	(0.027)	–	–	–	–
Borrowing skills	–	–	–	–	0.342*	0.077	0.031	0.216**
	–	–	–	–	(0.199)	(0.176)	(0.265)	(0.094)
	–	–	–	–	–	–	–	–
Age	1.513***	1.741***	1.317***	1.417***	1.438***	1.647***	1.167***	1.325***
	(0.3)	(0.234)	(0.194)	(0.116)	(0.296)	(0.223)	(0.181)	(0.111)
Marital status	0.269	-0.168	-0.122	0.029	0.275*	-0.147	-0.075	0.064
	(0.168)	(0.129)	(0.14)	(0.083)	(0.164)	(0.129)	(0.138)	(0.082)
Gender	-0.051	0.045	-0.132	-0.161*	0.011	0.076	-0.106	-0.119
	(0.173)	(0.14)	(0.14)	(0.083)	(0.166)	(0.137)	(0.137)	(0.082)
Education level	1.494***	.743***	1.123***	1.012***	1.621***	0.798***	1.185***	1.075***
	(0.195)	(0.148)	(0.209)	(0.083)	(0.189)	(0.147)	(0.207)	(0.082)
Income	0.034	-0.008	-0.008	0.034	0.005	-0.006	-0.007	0.035
	(0.056)	(0.039)	(0.048)	(0.025)	(0.056)	(0.039)	(0.048)	(0.025)
_cons	-7.527***	-8.745***	-6.493***	-7.062***	-6.776***	-8.467***	-5.997***	-6.785***
	(1.414)	(1.134)	(1.004)	(.584)	(1.376)	(1.114)	(0.994)	(0.574)
Region dummies	YES	YES	YES	YES	YES	YES	YES	YES
Observations	484	1888	700	3293	484	1888	700	3293
Pseudo R^2	0.296	0.175	0.15	0.175	0.278	0.169	0.139	0.17

Note: Robust standard errors in parentheses. ***, **, * denote significance at 1%, 5%, and 10%, respectively. The en dash (–) indicates not applicable.

4.3.3 Heterogeneous on household age categories

The study further tested the robustness of the findings by examining the relationship between health insurance enrolment and financial literacy, based on how numeracy and borrowing skills interact with health insurance across different household age groups, along with other control variables. Table 8 suggests that numeracy and borrowing skills were associated with enrolment among non-elderly Hhs but not among elderly Hhs. One possible reason is that younger, working-age adults may use financial knowledge more actively because they have longer planning horizons, greater exposure to education and digital or formal financial services, and more opportunities to compare and purchase insurance. Among older households, enrolment may depend more on affordability, health status, family support, eligibility conditions, trust, or previous experience with providers. These constraints could weaken the conversion of financial knowledge into action. Borrowing skills again display a less stable relationship because credit knowledge is not equivalent to insurance literacy and may facilitate enrolment only when households have access to credit and sufficient repayment capacity. The age-specific findings therefore supported targeted communication and simplified enrolment for older households, while retaining numeracy and insurance education for younger groups. The proposed mechanisms are plausible interpretations and require direct testing in future research.

4.3.4 Financial literacy mechanism affecting health insurance enrolment

The study also examined the mechanisms through which financial literacy could influence insurance consumption. Table 9 identifies three channels as dependent variables in the analysis: insurance works, which measures the effectiveness of the insurance in place; insurance benefits, which represent the benefits or perceived value of insurance; and insurance costs, which refer to the costs associated with insurance. Each outcome is

modelled using a similar set of independent variables, including region dummies.

Table 8. Heterogeneous on household age categories

Probit Model	(1)	(2)	(3)	(4)
	Dependent Variable: Health Insurance			
	Numeracy Skills		Borrowing Skills	
	Non-Elderly	Elderly	Elderly	Non-Elderly
Numeracy skills	0.155**	0.023	–	–
	(0.077)	(0.02)	–	–
Borrowing skills	–	–	0.104	0.196**
	–	–	(0.199)	(0.088)
Marital status	0.057	-0.049	-0.03	0.105
	(0.113)	(0.073)	(.131)	(0.069)
Gender	0.164	-0.252***	-.15	-.207***
	(0.126)	(0.074)	(0.131)	(0.069)
Education level	0.772***	1.032***	1.2***	0.733***
	(0.114)	(0.082)	(0.158)	(0.064)
Employment status	0.986***	0.752***	0.814**	0.964***
	(0.115)	(0.092)	(0.323)	(0.073)
Income	0.019	0.03	0.014	0.028
	(0.039)	(0.022)	(.038)	(0.022)
_cons	-2.508***	-1.535***	-0.988*	-2.023***
	(0.575)	(0.334)	(0.554)	(0.345)
Region dummies	YES	YES	YES	YES
Observations	4,193	1,022	1,022	4,193
Pseudo R^2	0.255	0.148	0.123	0.195

Note: The variable Elderly is coded as 1 if the heads of household (Hhs) is aged 60 years or older, and 0 if the Hh is between 16 and 59 years old. This threshold was selected as a benchmark because 60 years is the official mandatory retirement age under the Tanzanian employment policy. Robust standard errors in parentheses. ***, **, * denote significance at 1%, 5%, and 10%, respectively. The en dash (–) indicates not applicable.

Table 9. Mechanism through which financial literacy affects health insurance enrolment

Probit Model	(1)	(2)	(3)
Dependent Variable	Insurance Works	Insurance Benefits	Insurance Costs
Numeracy skills	0.126***	-0.052***	-0.042***
	(0.013)	(0.019)	(0.014)
Age	0.225***	-0.061	-0.006
	(0.055)	(0.086)	(0.059)
Marital status	0.045	0.041	-0.181***
	(0.044)	(0.066)	(0.047)
Gender	-0.015	0.018	-0.062
	(0.045)	(0.067)	(0.048)
Education level	0.566***	-0.265***	-0.07
	(0.053)	(0.087)	(0.056)
Employment status	0.366***	-0.1	-0.158**
	(0.07)	(0.127)	(0.078)
Income	0.018	-0.014	0.012
	(0.014)	(0.02)	(0.014)
_cons	-1.291***	-0.65	0.216
	(0.294)	(0.451)	(0.314)
Region dummies	YES	YES	YES
Observations	5230	4681	4681
Pseudo R^2	0.082	0.076	0.039

Note: Insurance Works is 1 if the head of household (Hh) understands how insurance works; otherwise, it is 0. Insurance Benefits is 1 if the Hh knows the benefits of health insurance and 0 if otherwise. Insurance Costs is 1 if the Hh could not afford insurance service; otherwise, it is 0. Other variables are as defined in Table 1. Robust standard errors (SEs) are in parentheses. ***, **, * denote significance at 1%, 5%, and 10%, respectively.

In column 1, the dependent variable is how insurance works. The results showed a positive and significant coefficient of 0.126 for numeracy skills at the 1% significance level. This means Hh with higher numeracy skills are more likely to understand how health insurance works. With higher numeracy skills, households can better interpret the intricacies of insurance policies and related premium and risk-pooling calculations.

In column 2, the dependent variable is insurance benefit. The results showed a negative coefficient of -0.052 for numeracy skills at the 1% significance level. Higher numeracy skills were associated with lower health insurance

enrolment, based on insurance benefits. This finding can be interpreted as follows: households with higher numeracy skills recognize that insurance policies involve a lot of information and therefore tend not to exaggerate their knowledge. Alternatively, households with lower numeracy skills can have an exaggerated sense of knowledge due to subjective bias.

In column 3, the dependent variable is insurance cost. The results recorded a negative coefficient of -0.042 for numeracy skills at the 1% significance level. A negative coefficient means that higher financial literacy is associated with lower insurance costs. Higher numeracy skills greatly decrease the probability that households claim that they cannot afford health insurance. It can be assumed that people with higher numeracy skills manage their family budgets better and can weigh risks and benefits when buying insurance.

4.3.5 Endogeneity check in the effect of financial literacy on health insurance enrolment

One potential source of endogeneity in estimating the effect of financial literacy on household health-insurance enrolment is that financial literacy may be correlated with unobserved characteristics that also influence insurance decisions, such as financial decision-making capacity. To address this concern, the study used a two-stage least squares (2SLS) instrumental-variable approach, using a binary indicator equal to one if household members are involved in financial decisions about purchasing goods and services, and zero if otherwise, as an instrument for financial literacy. Table 10 presents the results of a 2SLS estimation used to address potential endogeneity in the key variable, financial literacy (numeracy skills), when examining its effect on enrolment into health Insurance. Column 1 presents the first-stage regression that estimates numeracy skills using instrumental variables for financial decision-making and controls. For involvement in financial decision-making, the result showed a positive coefficient of 0.252 at the 1% significance level. This means that involvement in financial decisions, such as purchasing goods and services, spending and saving money, increases households' numeracy skills and improves financial literacy.

Table 10. Endogeneity check in the effect of financial literacy on health insurance enrolment

Probit Model	**(1)** **1st Stage**	**(2)** **2nd Stage**
Dependent Variable	**Numeracy Skills**	**Health Insurance**
Numeracy skills	–	0.21***
	–	(0.028)
Financial decisions involvement	0.252***	–
	(0.1)	–
Age	-0.907***	0.388***
	(0.061)	(0.03)
Marital status	0.26***	-0.051***
	(0.046)	(0.016)
Gender	0.368***	-0.106***
	(0.048)	(0.018)
Education level	0.584***	0.048*
	(0.043)	(0.025)
Employment status	0.141***	0.192***
	(0.054)	(0.024)
Income	0.019	-0.001
	(0.014)	(0.004)
_cons	2.318***	-1.298***
	(0.321)	(0.113)
Region dummies	YES	YES
Observations	5230	5230
R^2	0.206	0.526
1st stage *F*-statistic	–	87.84

Note: Financial decisions involvement is 1 if Hh is involved in financial decisions; otherwise, it is 0. Other variables are defined in Table 1. Robust standard errors in parentheses. ***, **, * denote significance at 1%, 5%, and 10%, respectively. The en dash (–) indicates not applicable.

Meanwhile, the results showed that older Hh had lower financial literacy, while married, male, and employed Hh tended to have higher financial literacy. Hh with higher education had higher financial literacy, suggesting that higher education improved financial literacy and that employed individuals had higher financial literacy. First stage *F*-statistic = 87.84: A high *F*- statistic indicated that the instruments used were potentially strong predictors of numeracy skills in respect of financial literacy, thus fulfilling a crucial requirement for 2SLS estimation.

Column 2 reports the second-stage regression results. This stage used instrumented numeracy skills to estimate its effect on enrolment in health insurance, while using the same control variables. For numeracy skills, the results showed a positive coefficient of 0.21 at the 1% significance level. This indicated that a one-unit increase in

instrumented numeracy skills was associated with a 0.21-unit increase in the health insurance outcome. This suggested that higher financial literacy led to desirable health insurance outcomes through improved decision-making and a deeper understanding of insurance options. The instrument's validity was supported by a strong first-stage *F*-statistic and Hansen *J*-statistic, suggesting that the endogeneity problem in numeracy skills was effectively addressed.

5. Conclusions

This study investigated the relationship between household financial literacy and health insurance consumption in Tanzania, to address research gaps identified in recent policy reports and academic literature. Our analysis of nationally representative data revealed three major findings: first, financial literacy significantly increased the probability of health insurance enrolment, with effects ranging from 9% (numeracy skills) to 22% (borrowing skills). Second, the interrelationship operated primarily through improved understanding of insurance mechanics and reduced cost barriers. Third, effects were heterogeneous, while stronger in urban areas, among formally employed individuals, and within younger households.

These findings extended financial literacy theory to the health insurance context in low-income settings, showing that financial literacy affected not only traditional financial behaviours (saving and investing) but also health-related financial protection decisions. The heterogeneous effects aligned with behavioural economics perspectives and suggested that structural constraints mediated the translation of knowledge into behaviour. With only 11.9% of Tanzanian households currently insured, our results underscored that financial literacy interventions could meaningfully increase coverage. However, standalone education programs might have limited impact unless integrated with affordability measures (premium subsidies), structural improvements (rural service delivery), and trust-building initiatives. Through a governance and risk-management lens, these findings position health insurance as more than a healthcare-financing product: it is a household mechanism for pooling risk and limiting exposure to unpredictable medical expenditure.

Financial literacy represents a modifiable determinant that, when coupled with structural reforms, could accelerate progress toward universal health coverage and financial protection for vulnerable households. Financial literacy could strengthen demand-side governance by enabling households to evaluate premiums, benefits, exclusions, and provider reliability. However, effective protection also requires transparent scheme rules, accountable administration, accessible complaint and claims procedures, and dependable service delivery. Evidence from low- and middle-income countries shows that financial-protection effects from insurance vary by context and scheme design. Offices such as the Tanzania Insurance Regulatory Authority should coordinate literacy initiatives with affordability measures, improvements of rural access, and institutional safeguards rather than treating them as standalone solutions. Such integration would make insurance uptake more likely to translate into genuine resilience against medical expenditure shocks and reduced household financial vulnerability.

This study has few limitations. The cross-sectional design prevented analysis of literacy and insurance dynamics over time. While instrumental variables addressed endogeneity, unobserved factors may persist. Financial literacy measures, though based on established frameworks, might not capture all relevant dimensions in the Tanzanian context. Improving health insurance coverage in Tanzania requires addressing both supply-side constraints and demand-side behavioural barriers. Future studies should employ longitudinal designs to track behavioural change, experimental approaches to test integrated literacy-insurance interventions, and mixed methods to explore cultural and trust factors. Research should also examine the role of digital financial literacy as mobile insurance platforms expand.

Author Contributions

Conceptualization, H.M.; methodology, F.M.; software, F.M.; validation, H.M. and F.M.; formal analysis, F.M.; writing—original draft preparation, H.M.; writing—review and editing, H.M.; visualization, F.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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